CALCULUS

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1. Real sequences

Definition. *The function $x : \mathbb{N} \to \mathbb{R}$ (where the domain is the set of positive integers and the range is the set of real numbers) is called a real sequence.*

Denotions:

Let $n \in \mathbb{N}$ be arbitrary, then instead of $x(n)$ we write x_n, for the sequence we use the $(x_n)_{n\in\mathbb{N}}$ notion. The real number x_n is called the n-th element of the sequence $(x_n)_{n\in\mathbb{N}}$.

Definition. *The real sequence $(x_n)_{n\in\mathbb{N}}$ is called bounded below, if there exists $k \in \mathbb{R}$ (lower bound), such that*

$$k \le x_n \qquad (\forall\, n \in \mathbb{N}) .$$

The real sequence $(x_n)_{n\in\mathbb{N}}$ is called bounded above, if there exists $K \in \mathbb{R}$ (upper bound), such that

$$x_n \le K \qquad (\forall\, n \in \mathbb{N}) .$$

The real sequence $(x_n)_{n\in\mathbb{N}}$ is called bounded, if it is bounded above and below.

Definition.
The real sequence $(x_n)_{n\in\mathbb{N}}$ is called increasing, if for all $n \in \mathbb{N}$ the inequality $x_n \le x_{n+1}$ holds.
The real sequence $(x_n)_{n\in\mathbb{N}}$ is called decreasing, if for all $n \in \mathbb{N}$ the inequality $x_n \ge x_{n+1}$ holds.
If the above inequalities are strict for all $n \in \mathbb{N}$, then we call them strictly increasing/decreasing.
If a sequence is increasing or decreasing then we call it monotonic.

Remark.

(i) Increasing real sequences are bounded below.
(ii) Decreasing real sequences are bounded above.

Definition. *The real sequence $(x_n)_{n\in\mathbb{N}}$ is convergent, if there exists an $x \in \mathbb{R}$, such that for all $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$, such that for all $n \ge n(\varepsilon)$ ($n \in \mathbb{N}$) the inequality $|x_n - x| < \varepsilon$ holds. The number x is called the limit of the sequence $(x_n)_{n\in\mathbb{N}}$. Denote it with*

$$\lim_{n\to\infty} x_n = x$$

.

Definition. *The real sequence $(x_n)_{n\in\mathbb{N}}$ is divergent if it is not convergent.*

Definition. *The real sequence $(x_n)_{n\in\mathbb{N}}$ diverges to $+\infty$, if for all $K \in \mathbb{R}$ there exists number $N(K) > 0$ such that if $n \ge N(K)$, then $x_n > K$. We denote it with*

$$\lim_{n\to\infty} x_n = +\infty.$$

The real sequence $(x_n)_{n\in\mathbb{N}}$ diverges to $-\infty$, if for all $k \in \mathbb{R}$ there exists number $N(k) > 0$ such that if $n \ge N(k)$, then $x_n < k$. We denote it with

$$\lim_{n\to\infty} x_n = -\infty.$$

Definition. *Let*

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\} .$$

Theorem. ***(unique limit)***
If the sequence $(x_n)_{n\in\mathbb{N}}$ converges to both x and y, then $x = y$.

Theorem. ***(limit and boundedness)***
If the sequence $(x_n)_{n\in\mathbb{N}}$ is convergent then it is bounded.

Remark. *$\alpha \in \mathbb{R}$ is called the least upper bound of the set $B \subset \mathbb{R}$, if*

- *α is upper bound of B (so α is greater than or equal to all elements of B)*
- *for any upper bound β of B $\alpha \leq \beta$ holds.*

If there exists the least upper bound of B, then we denote it with $\sup B$ (supremum B). Similarly we can define the greatest lower bound as well: $\inf B$ (infimum B).

Definition. *Let $D \subset \mathbb{R}$. $x_0 \in \mathbb{R}$ is a limit point of D if every neighbourhood of x_0 contains at least one point of D different from x_0 itself. The set of limit points of D is denoted by D'.*

Theorem. ***(limit and monotonicity)***
If the sequence $(x_n)_{n\in\mathbb{N}}$ is increasing and bounded above then it is convergent and

$$\lim_{n\to\infty} x_n = \sup\left\{x_n \,|\, n \in \mathbb{N}\right\}.$$

If the sequence $(x_n)_{n\in\mathbb{N}}$ is decreasing and bounded below then it is convergent and

$$\lim_{n\to\infty} x_n = \inf\left\{x_n \,|\, n \in \mathbb{N}\right\}.$$

Definition. *The sequence $(y_n)_{n\in\mathbb{N}}$ is called the subsequence of $(x_n)_{n\in\mathbb{N}}$ if there exists strictly increasing function $\varphi\colon \mathbb{N} \to \mathbb{N}$ such that for all $n \in \mathbb{N}$ $y_n = x_{\varphi(n)}$ holds.*

Theorem. ***(limit of a subsequence)*** *If the sequence $(x_n)_{n\in\mathbb{N}}$ is convergent and has the limit $x \in \overline{\mathbb{R}}$, then for any subsequence $(y_n)_{n\in\mathbb{N}}$ of $(x_n)_{n\in\mathbb{N}}$ the following is true:*

$$\lim_{n\to\infty} y_n = x.$$

Theorem. *All real sequence has a monotonic subsequence.*

Theorem. ***(Bolzano–Weierstrass)***
All real bounded sequence has a convergent subsequence.

Definition. *Let $(x_n)_{n\in\mathbb{N}}$ be a bounded sequence. Denote by X the set of limits of convergent subsequences. The (existing) numbers $\sup X$ and $\inf X$ are called the limes superior and limes inferior of the sequence $(x_n)_{n\in\mathbb{N}}$.*
Notation:

$$\limsup_{n\to\infty} x_n, \qquad \liminf_{n\to\infty} x_n.$$

If $(x_n)_{n\in\mathbb{N}}$ is not bounded above (or below) then $\limsup x_n = +\infty$ (and $\liminf x_n = -\infty$).

Remark. *For arbitrary sequence $(x_n)_{n\in\mathbb{N}}$:*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Theorem. *The real sequence $(x_n)_{n\in\mathbb{N}}$ is convergent if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n.$$

Definition. *The sequence $(x_n)_{n\in\mathbb{N}}$ is called Cauchy sequence, if for all $\varepsilon > 0$ there exists $N(\varepsilon) \in \mathbb{N}$, such that for all $n, m \in \mathbb{N}$, $n, m \geq N(\varepsilon)$ the following is true:*

$$|x_n - x_m| < \varepsilon.$$

(This means that the elements of the sequence are „close to each other”.)

Theorem. ***(Cauchy's convergence test for sequences)***
A real sequence is convergent if and only if it is a Cauchy sequence.

Theorem. ***(convergence and operations - sequences)***
Let $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ be sequences with

$$\lim_{n\to\infty} x_n = x$$

és

$$\lim_{n\to\infty} y_n = y,$$

$x, y \in \mathbb{R}$. Let $\lambda \in \mathbb{R}$ be arbitrary. Then

(i) the sequence $(x_n + y_n)_{n\in\mathbb{N}}$ is convergent and

$$\lim_{n\to\infty} (x_n + y_n) = x + y;$$

(ii) the sequence $(\lambda x_n)_{n\in\mathbb{N}}$ is convergent and

$$\lim_{n\to\infty} (\lambda x_n) = \lambda x;$$

(iii) the sequence $(x_n y_n)_{n\in\mathbb{N}}$ is convergent and

$$\lim_{n\to\infty} (x_n y_n) = xy;$$

(iv) if for all $n \in \mathbb{N} : y,\ y_n \neq 0$, then the sequence $\left(\frac{x_n}{y_n}\right)_{n\in\mathbb{N}}$ is convergent and

$$\lim_{n\to\infty} \left(\frac{x_n}{y_n}\right) = \frac{x}{y}.$$

Theorem. ***(convergence and order - sequences)***
Let $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ be sequences such that $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$, and

a) there exists $N \in \mathbb{N}$, such that for all $n > N$ $x_n \leq y_n$ is true, then $x \leq y$,
b) if $x < y$, then there exists $N \in \mathbb{N}$, such that for all $n > N$ $x_n < y_n$ is true.

Theorem. ***(sandwich theorem)*** *Let $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ and $(z_n)_{n\in\mathbb{N}}$ be sequences such that*

(i)

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n;$$

(ii) for all $n \in \mathbb{N}$:

$$x_n \leq z_n \leq y_n.$$

Then the sequence $(z_n)_{n\in\mathbb{N}}$ *is convergent and*

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} z_n = \lim_{n\to\infty} y_n.$$

Theorem. **(geometric sequence)** *Let* $a \in \mathbb{R}$ *and*

$$x_n = a^n \qquad (n \in \mathbb{N}).$$

Then

- *if* $|a| < 1$, *then the sequence* $(x_n)_{n\in\mathbb{N}}$ *is convergent and* $\lim_{n\to\infty} x_n = 0$*;*
- *if* $a = 1$, *then the sequence* $(x_n)_{n\in\mathbb{N}}$ *is convergent and its limit is* 1*;*
- *if* $a > 1$, *then the sequence* $(x_n)_{n\in\mathbb{N}}$ *tends to* $+\infty$*;*
- *if* $a \leq -1$, *then the sequence* $(x_n)_{n\in\mathbb{N}}$ *is divergent.*

Theorem. **(limit of basic sequences)**

- *Let* $k \in \mathbb{N}$ *be fixed. Then*
$$\lim_{n\to\infty} n^k = \infty,$$
$$\lim_{n\to\infty} \sqrt[k]{n} = \infty$$
and
$$\lim_{n\to\infty} n^p = \infty,$$
if $p = \frac{k}{l}$ $(k, l \in \mathbb{N})$.
- *Let* $a \in \mathbb{R}$, $a > 0$. *Then*
$$\lim_{n\to\infty} \sqrt[n]{a} = 1.$$
- $$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$
- *Let* $a \in \mathbb{R}$, $a > 1$. *Then*
$$\lim_{n\to\infty} \frac{a^n}{n!} = 0.$$
- $$\lim_{n\to\infty} \sqrt[n]{n!} = \infty.$$
- *If* $a > 1$, *then for all fixed* $k \in \mathbb{N}$*:*
$$\lim_{n\to\infty} \frac{n^k}{a^n} = 0.$$
- *The sequence* $\left(\left(1 + \frac{1}{n}\right)^n\right)_{n\in\mathbb{N}}$ *is convergent. Its limit is denoted by* e.
- *Let* $(p_n)_{n\in\mathbb{N}}$ *be a sequence with limit* $+\infty$ *or* $-\infty$ *and*
$$x_n = \left(1 + \frac{1}{p_n}\right)^{p_n} \qquad (n \in \mathbb{N})$$
Then
$$\lim_{n\to\infty} \left(1 + \frac{1}{p_n}\right)^{p_n} = e.$$

Definition. $D \subset \mathbb{R}$ *is compact if and only if it is closed and bounded.*

2. Real series

Definition. *Let $(x_n)_{n\in\mathbb{N}}$ be a real sequence and*

$$S_1 = x_1, \qquad S_n = x_1 + \cdots + x_n \quad (n \in \mathbb{N}, n \geq 2)\,.$$

$(S_n)_{n\in\mathbb{N}}$ is called the partial sum of the sequence $\sum x_n$. If the sequence $(S_n)_{n\in\mathbb{N}}$ has a limit, then the value

$$\sum_{n=1}^{\infty} x_n = \lim_{n\to\infty} S_n$$

is called the sum of the real series and we say that the series $\sum_{n=1}^{\infty} x_n$ is convergent.

If it is not convergent, then we call it divergent.

Definition. *Let $\sum_{n=1}^{\infty} x_n$ be a real series. If the series $\sum_{n=1}^{\infty} |x_n|$ is convergent, then we say that the series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent. If it is convergent, but not absolutely convergent, we call it conditionally convergent.*

Theorem. ***(Cauchy's convergence test for series)***

The series $\sum_{n=1}^{\infty} x_n$ is convergent if and only if for all $\varepsilon > 0$ there exist $N(\varepsilon) \in \mathbb{N}$ such that

$$\left|\sum_{k=n+1}^{m} x_k\right| < \varepsilon$$

holds if $n, m \geq N(\varepsilon),\ m > n,\ m, n \in \mathbb{N}$.

Theorem. ***(connection of conv. and absolutely conv. series)***

If a real series is absolutely convergent, then it is also convergent.

Theorem. ***(necessary condition for convergence)***

If the series $\sum_{n=1}^{\infty} x_n$ is convergent, then $\lim_{n\to\infty} x_n = 0$.

Examples:

Let $a \in \mathbb{R},\ |a| < 1$. Then

$$\sum_{n=1}^{\infty} a^n$$

geometric series is convergent and

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(a + a^2 + \cdots + a^n\right) = \lim_{n\to\infty} a \cdot \frac{a^n - 1}{a - 1} = \frac{a}{1-a}.$$

The $\sum_{n=1}^{\infty} \frac{1}{n}$ harmonic series is divergent.

Theorem. ***(convergence and operations)***

If the series $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are convergent, $\lambda \in \mathbb{R}$, then

$$\sum_{n=1}^{\infty} (x_n + y_n) \quad \text{and} \quad \sum_{n=1}^{\infty} \lambda x_n$$

series are also convergent and

$$\sum_{n=1}^{\infty}(x_n + y_n) = \sum_{n=1}^{\infty} x_n + \sum_{n=1}^{\infty} y_n,$$

$$\sum_{n=1}^{\infty} \lambda x_n = \lambda \sum_{n=1}^{\infty} x_n,$$

and

$$\left| \sum_{n=1}^{\infty} x_n \right| \leq \sum_{n=1}^{\infty} |x_n| \, .$$

Theorem. ***(Comparison test)***

Let $\sum_{n=1}^{\infty} x_n$ *and* $\sum_{n=1}^{\infty} y_n$ *be series with nonnegative elements, such that* $x_n \leq y_n$ *holds for all* $n \in \mathbb{N}$*. Then*

- *if* $\sum_{n=1}^{\infty} y_n$ *is convergent, then* $\sum_{n=1}^{\infty} x_n$ *is also convergent;*
- *if* $\sum_{n=1}^{\infty} x_n$ *is divergent, then* $\sum_{n=1}^{\infty} y_n$ *is divergent.*

Theorem. ***(root test)***

Let $\sum_{n=1}^{\infty} x_n$ *be real series.*

- *If* $\limsup_{n\to\infty} \sqrt[n]{|x_n|} < 1$*, then the series* $\sum_{n=1}^{\infty} x_n$ *is absolutely convergent;*
- *If* $\liminf_{n\to\infty} \sqrt[n]{|x_n|} > 1$*, then the series* $\sum_{n=1}^{\infty} x_n$ *is divergent.*

Theorem. ***(ratio test)***

Let $\sum_{n=1}^{\infty} x_n$ *be real series, for which* $x_n \neq 0$ *for all* $n \in \mathbb{N}$*..*

- *If* $\limsup_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} < 1$*, then the series* $\sum_{n=1}^{\infty} x_n$ *is absolutely convergent;*
- *If* $\liminf_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} > 1$*, then the series* $\sum_{n=1}^{\infty} x_n$ *is divergent.*

3. Real functions

Definition. *Let $D \subset \mathbb{R}$ be nonempty set. The function $f\colon D \to \mathbb{R}$ is called a real function.*

Definition. *Let $D \subset \mathbb{R}$ be nonempty set and $f\colon D \to \mathbb{R}$ be a real function. The function f is continuous at point $x_0 \in D$, if for all $\varepsilon > 0$ there exists $\delta > 0$, such that if $x \in D$ satisfies $|x - x_0| < \delta$, then*

$$|f(x) - f(x_0)| < \varepsilon.$$

If the function f is continuous at every point of the set D, then we say that f is continuous on the set D.

Theorem. ***(connection with the limits of sequences)***
Let $D \subset \mathbb{R}$ be nonempty set, $f\colon D \to \mathbb{R}$. The function f is continuous at the point $x_0 \in D$ if and only if for all sequences $(x_n)_{n\in\mathbb{N}}$ from set D, which converges to x_0, the sequence $(f(x_n))_{n\in\mathbb{N}}$ tends to $f(x_0)$.

Theorem. ***(continuity and operations)***
Let $D \subset \mathbb{R}$ be a nonempty set. If the functions $f, g : D \to \mathbb{R}$ are continuous at point $x_0 \in D$, then

(i) the function $f + g$ is continuous at x_0;
(ii) for any $\lambda \in \mathbb{R}$ the function λf is continuous at x_0;
(iii) the function $f \cdot g$ is continuous at x_0;
(iv) if for any $x \in D$: $g(x) \neq 0$, then the function $\dfrac{f}{g}$ is continuous at x_0.

Theorem. ***(continuity of composite functions)***
Let $D \subset \mathbb{R}$ be a nonempty set and $f : D \to \mathbb{R}$ and $g : f(D) \to \mathbb{R}$ be real functions. If the function f is continuous at $x_0 \in D$ and the function g is continuous at $f(x_0) \in f(D)$, then the function $g \circ f$ is continuous at x_0.

Theorem. ***(continuity and compactness)***
Let $D \subset \mathbb{R}$ be compact set, $f\colon D \to \mathbb{R}$ be continuous function. Then the set $f(D) \subset \mathbb{R}$ is compact.

Theorem. ***(Bolzano theorem)***
Let $a, b \in \mathbb{R}$, $a < b$, $f\colon [a, b] \to \mathbb{R}$ be continuous function. If $f(a) < f(b)$ and $\eta \in \mathbb{R}$ is so that $f(a) < \eta < f(b)$, then there exists $\xi \in (a, b)$ such that $f(\xi) = \eta$.

Definition. *Let $D \subset \mathbb{R}$ nonempty set, $f\colon D \to \mathbb{R}$. The function f is called uniformly continuous on the set D, if for all $\varepsilon > 0$ there exists $\delta > 0$ such that if $x, y \in D$ satisfy $|x - y| < \delta$, then*

$$|f(x) - f(y)| < \varepsilon.$$

Remark. *If the function f is uniformly continuous on the set D, then it is also continuous at every point of D.*

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $f\colon D \to \mathbb{R}$ be real function. If the set f is not continuous at a point $x_0 \in D$ then we say that the point $x_0 \in D$ is a discontinuity of the function f.*

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $f : D \to \mathbb{R}$, $x_0 \in D'$ and $A \in \mathbb{R}$. The function f has limit A at point x_0, if for all $\varepsilon > 0$ there exists $\delta > 0$, such that if $x \in D$ and $|x - x_0| < \delta$, then $|f(x) - A| < \varepsilon$. We denote it with*

$$\lim_{x \to x_0} f(x) = A.$$

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $f : D \to \mathbb{R}$, $x_0 \in D'$. The limit of f at x_0 is $+\infty$ (or $-\infty$), if for all $K \in \mathbb{R}$ (or $k \in \mathbb{R}$) there exists $\delta > 0$, such that if $x \in D$ and $|x - x_0| < \delta$, then $f(x) > K$ (or $f(x) < k$). We denote these with $\lim_{x \to x_0} f(x) = +\infty$ (or $\lim_{x \to x_0} f(x) = -\infty$).*

Definition. *Let $D \subset \mathbb{R}$ nonempty set, which is not bounded above (or below), $f : D \to \mathbb{R}$. The limit of the function f at $+\infty$ (or at $-\infty$) is A, if for all $\varepsilon > 0$ there exists $K \in \mathbb{R}$ (or $k \in \mathbb{R}$), such that if $x \in D$ and $x \geq K$ (or $x \leq k$), then $|f(x) - A| < \varepsilon$. We denote these with $\lim_{x \to +\infty} f(x) = A$ (or $\lim_{x \to -\infty} f(x) = A$).*

Remark. *One can define the infinite limits of f at $+\infty$ (or $-\infty$) as well.*

Theorem. ***(connection with the limits of sequences)***
Let $D \subset \mathbb{R}$ be nonempty set, $f : D \to \mathbb{R}$, $x_0 \in D'$ and $A \in \overline{\mathbb{R}}$. $\lim_{x \to x_0} f(x) = A$ is true if and only if for all sequences $(x_n)_{n \in \mathbb{N}}$ from D which tends to x_0 the following is true: $\lim_{n \to \infty} f(x_n) = A$.

Theorem. ***(limit and continuity)***
Let $D \subset \mathbb{R}$ be nonempty set, $f : D \to \mathbb{R}$ and $x_0 \in D$. The function f is continuous at x_0 if and only there exists the limit $\lim_{x \to x_0} f(x)$ and

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Theorem. ***(limit and operations)***
Let $D \subset \mathbb{R}$ be nonempty set, $x_0 \in D'$, $f, g : D \to \mathbb{R}$ and $A, B \in \mathbb{R}$. If the functions f and g has limit at point x_0 and

$$\lim_{x \to x_0} f(x) = A \quad \textit{and} \quad \lim_{x \to x_0} g(x) = B,$$

then

(i) the function $f + g$ has limit at point x_0 and

$$\lim_{x \to x_0} (f(x) + g(x)) = A + B;$$

(ii) for all $\lambda \in \mathbb{R}$ the function $\lambda \cdot f$ has limit at x_0 and

$$\lim_{x \to x_0} \lambda \cdot f(x) = \lambda \cdot A;$$

(iii) the function $f \cdot g$ has limit at x_0 and

$$\lim_{x \to x_0} f(x) \cdot g(x) = A \cdot B$$

(iv) if $B \neq 0$ and $g(x) \neq 0$ is satisfied for all $x \in D$, then the function $\dfrac{f}{g}$ has limit at x_0 and

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{A}{B}.$$

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $x_0 \in D'$ $f : D \to \mathbb{R}$. The function f has the right- (or left-) hand side limit at point x_0, if there exists such $A \in \mathbb{R}$, so that for all $\varepsilon > 0$ there exists $\delta > 0$ such that if $x \in D$: $x_0 < x < x_0 + \delta$ (or $x_0 - \delta < x < x_0$), then $|f(x) - A| < \varepsilon$.*

We use the notation $\lim_{x \to x_0+0} f(x) = A$ (or $\lim_{x \to x_0-0} f(x) = A$).

Theorem. *Let $D \subset \mathbb{R}$ be nonempty set, $x_0 \in D'$ $f : D \to \mathbb{R}$. If the function f has the limit at point x_0, then the function f has the right- and left-hand side limit at x_0 and*

$$\lim_{x \to x_0+0} f(x) = \lim_{x \to x_0-0} f(x) = \lim_{x \to x_0} f(x)$$

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $f_n \colon D \to \mathbb{R}$, $(n \in \mathbb{N})$ be functions. The sequence of $(f_n)_{n \in \mathbb{N}}$ is called sequence of functions and if $S_n = f_1 + \cdots + f_n$ $(n \in \mathbb{N})$, then $(S_n)_{n \in \mathbb{N}}$ is called series of functions. Denotion: $\sum_{n=1}^{\infty} f_n$.*

Definition. *The sequence of functions $(f_n)_{n \in \mathbb{N}}$ is pointwise convergent in $x \in D$, if the real sequence $(f_n(x))_{n \in \mathbb{N}}$ is convergent. If this is true in all of the points of set D, then we say that the sequence $(f_n)_{n \in \mathbb{N}}$ is pointwise convergent on the set D.*

Then the function

$$f(x) = \lim_{n \to \infty} f_n(x) \qquad x \in D$$

is called the pointwise limit function.

Definition. *Let $D \subset \mathbb{R}$ be nonempty set, $f_n \colon D \to \mathbb{R}$, $(n \in \mathbb{N})$ be a sequence of functions, which converges pointwise on the set D and denote its limit function with f. The sequence of functions $(f_n)_{n \in \mathbb{N}}$ is uniformly convergent on set D, if for all $\varepsilon > 0$ there exists $N > 0$, such that $n \in \mathbb{N}$, $n > N$, then*

$$|f_n(x) - f(x)| < \varepsilon$$

is valid for all $x \in D$.

Remark. *Uniform convergence $\Longrightarrow$ pointwise convergence*

Theorem. ***(Weierstrass theorem)***

Let $[a, b] \subset \mathbb{R}$, $f \colon [a, b] \to \mathbb{R}$ be continuous function and $\varepsilon > 0$. Then there exist real polinomial P, such that

$$|f(x) - P(x)| < \varepsilon$$

holds for all $x \in [a, b]$.

Definition. *Let $(a_n)_{n \in \mathbb{N}}$ be real sequence, $x_0 \in \mathbb{R}$ and*

$$f_n(x) = \begin{cases} a_0, & \text{if } n = 0 \\ a_n(x - x_0)^n, & \text{if } n \neq 0. \end{cases}$$

Then the function series made from the sequence of functions $(f_n)_{n \in \mathbb{N} \cup \{0\}}$ is called power series. Elements of the sequence $(a_n)_{n \in \mathbb{N} \cup \{0\}}$ are called the coefficients, x_0 is called the center of the series. We can denote it with

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n.$$

Theorem. ***(Cauchy–Hadamard)*** *Let $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ be a power series and*

$$\alpha = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

Then

(i) If $\alpha = 0$, then the power series is convergent for all $x \in \mathbb{R}$;

(ii) If $0 < \alpha < +\infty$, then the power series is convergent in points $x \in \mathbb{R}$, where

$$\alpha\,|x-x_0| < 1$$

and it is divergent in points $x \in \mathbb{R}$, where $\alpha\,|x-x_0| > 1$;

(iii) If $\alpha = +\infty$, then the power series is convergent just at the point x_0.

Definition.

$$\rho = \begin{cases} +\infty, & \text{ha } \alpha = 0 \\ \dfrac{1}{\alpha}, & \text{if } 0 < \alpha < +\infty \\ 0, & \text{if } \alpha = +\infty. \end{cases}$$

ρ is called the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n(x-x_0)^n$.

Definition. *The function $\exp\colon \mathbb{R} \to \mathbb{R}$ defined by*

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \qquad (x \in \mathbb{R})$$

is called the exponential function.

Theorem. *The exponential function satisfies the following:*

(i)

$$\exp(0) = 1;$$

(ii)

$$\exp(x+y) = \exp(x)\cdot\exp(y) \qquad (x, y \in \mathbb{R})\,;$$

(iii)

$$\exp(-x) = \frac{1}{\exp(x)} \qquad (x \in \mathbb{R})\,.$$

Theorem. *The exponential functon is*

(i) continuous;

(ii) strictly increasing.

(iii) Its range is the set of positve real numbers;

(iv)

$$\lim_{x\to+\infty} \exp(x) = +\infty \quad \text{and} \quad \lim_{x\to-\infty} \exp(x) = 0$$

Theorem.

$$\exp(1) = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

Definition. *The exponential function is strictly increasing, so it is invertible. This inverse function is called the logarithmic function, and we denote it with* ln.

So the logarithmic function $\ln\colon]0,+\infty[\to \mathbb{R}$ satisfies

$$\exp(\ln(x)) = x \qquad (x\in(0,+\infty)).$$

Theorem. *The function* $\ln\colon (0,+\infty)\to\mathbb{R}$ *satisfies the following:*

(i)

$$\ln(1) = 0;$$

(ii)

$$\ln(x\cdot y) = \ln(x)+\ln(y) \qquad (x,y\in(0,+\infty));$$

(iii)

$$\ln\left(\frac{1}{x}\right) = -\ln(x) \qquad (x\in(0,+\infty));$$

(iv) *continuous;*
(v) *strictly increasing;*
(vi)

$$\lim_{x\to+\infty}\ln(x) = +\infty \quad \textit{and} \quad \lim_{x\to 0+}\ln(x) = -\infty.$$

Definition. *The functions*

$$\cosh(x) = \sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!} \qquad (x\in\mathbb{R}),$$

and

$$\sinh(x) = \sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!} \qquad (x\in\mathbb{R})$$

are called cosinus hiperbolicus and sinus hiperbolicus.

Theorem. *For any* $x\in\mathbb{R}$*:*

$$\cosh(x) = \frac{\exp(x)+\exp(-x)}{2} \qquad \sinh(x) = \frac{\exp(x)-\exp(-x)}{2}.$$

Theorem. *The functions cosinus hiperbolicus and sinus hiperbolicus satisfy the following*

(i) *they are continuous;*
(ii) $\cosh(0) = 1$ *and* $\sinh(0) = 0$;
(iii) $\cosh(x+y) = \cosh(x)\cosh(x)+\sinh(x)\sinh(x) \qquad (x,y\in\mathbb{R});$
(iv) $\sinh(x+y) = \sinh(x)\cosh(y)+\sinh(y)\cosh(x) \qquad (x,y\in\mathbb{R})$
(v) $\cosh(x) = \cosh(-x)$ *and* $\sinh(x) = -\sinh(x) \qquad (x\in\mathbb{R})$
(vi) $\cosh^2(x)-\sinh^2(x) = 1 \qquad (x\in\mathbb{R}).$

Definition. *The functions*

$$\tanh(x) = \frac{\sinh(x)}{\cos(x)} \qquad (x \in \mathbb{R})\,,$$

and

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} \qquad (x \in \mathbb{R} \setminus \{0\})$$

are called tangent hiperbolicus and cotangens hiperbolicus.

Definition. *The functions*

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \qquad (x \in \mathbb{R})\,,$$

and

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \qquad (x \in \mathbb{R})$$

are called cosinus and sinus.

Theorem. *(i)*

$$\cos(0) = 1 \quad \textit{and} \quad \sin(0) = 0;$$

(ii)

$$\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y) \qquad (x, y \in \mathbb{R})$$

(iii)

$$\sin(x+y) = \sin(x)\cos(y) + \sin(y)\cos(x) \qquad (x, y \in \mathbb{R})\,;$$

(iv)

$$\cos(-x) = \cos(x) \quad \textit{and} \quad \sin(-x) = -\sin(x) \qquad (x \in \mathbb{R})\,;$$

(v)

$$\sin^2(x) + \cos^2(x) = 1 \qquad (x \in \mathbb{R})\,.$$

Definition. *The functions*

$$\operatorname{tg}(x) = \frac{\sin(x)}{\cos(x)} \qquad \left(x \in \mathbb{R}, x \neq (2k+1)\frac{\pi}{2}\right),$$

and

$$\operatorname{ctg}(x) = \frac{\cos(x)}{\sin(x)} \qquad (x \in \mathbb{R}, x \neq k\pi)$$

are called tangent and cotangent.

Definition. *Let $D \subset \mathbb{R}$ be nonempty open set, $f\colon D \to \mathbb{R}$ be real function. The function φ defined by*

$$\varphi(x, x_0) = \frac{f(x) - f(x_0)}{x - x_0} \qquad (x, x_0 \in D,\ x \neq x_0)$$

is called the difference quotient function of f at x and x_0.

Definition. *Let $D \subset \mathbb{R}$ be nonempty open set and $x_0 \in D$. The function $f\colon D \to \mathbb{R}$ is differentiable at the point $x_0 \in D$ if the (finite) limit*

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

exists. $f'(x_0)$ is one of several common notations for the derivative at point x_0.

Theorem. ***(connection of continuity and differentiation)*** *Let $D \subset \mathbb{R}$ nonempty open set, $f\colon D \to \mathbb{R}$ and $x_0 \in D$. If the function f is differentiable at x_0, then it is also continuous at that point.*

Theorem. ***(differentiation and operations)*** *Let $D \subset \mathbb{R}$ nonempty open set, $x_0 \in D$ and let $f, g\colon D \to \mathbb{R}$ be differentiable at point x_0. Then*

- *the function $f + g$ is differentiable at x_0 and*
$$(f+g)'(x_0) = f'(x_0) + g'(x_0).$$
- *for arbitrary $\lambda \in \mathbb{R}$, the function $\lambda \cdot f$ is differentiable at point x_0 and*
$$(\lambda \cdot f)'(x_0) = \lambda \cdot f'(x_0).$$
- *the function $f \cdot g$ is differentiable at point x_0 and*
$$(f \cdot g)'(x_0) = f'(x_0) \cdot g(x_0) + f(x_0) \cdot g'(x_0).$$
- *if the point x_0 has a neighbourhood on which $g(x) \neq 0$, then the function $\dfrac{f}{g}$ is differentiable at point x_0 and*
$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}.$$

Theorem. ***(differentiation of the composite function)*** *Let $D \subset \mathbb{R}$ be nonempty open set, $x_0 \in D$ and let $g\colon D \to \mathbb{R}$, $f\colon g(D) \to \mathbb{R}$ be functions such that g is differentiable at point x_0 and f is differentiable at point $g(x_0)$. Then the function $f \circ g$ is differentiable at point x_0, moreover*

$$(f \circ g)'(x_0) = f'(g(x_0)) \cdot g'(x_0).$$

Theorem. ***(differentiation of the inverse function)***

Let $(a,b) \subset \mathbb{R}$ nonempty open interval, $f\colon (a,b) \to \mathbb{R}$ be contionuous and strictly increasing function. If f is differentiable at point $x_0 \in (a,b)$ and $f'(x_0) \neq 0$, then f^{-1} is differentiable at point $f(x_0) \in f((a,b))$ and

$$\left(f^{-1}\right)'(f(x_0)) = \frac{1}{f'(x_0)}.$$

Derivatives of basic functions

- if $n \in \mathbb{Z} \setminus \{0\}$ and $f(x) = x^n$, then $f'(x) = n \cdot x^{n-1}$;
- if $f(x) = \exp(x)$, then $f'(x) = \exp(x)$;
- if $f(x) = a^x$, then $f'(x) = a^x \cdot \ln(a)$;
- if $f(x) = \ln(x)$, then $f'(x) = \dfrac{1}{x}$;
- if $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$;

- if $f(x) = \sin(x)$, then $f'(x) = \cos(x)$;
- if $f(x) = \text{tg}(x)$, then $f'(x) = \dfrac{1}{\cos^2}(x)$;
- if $f(x) = \text{ctg}(x)$, then $f'(x) = -\dfrac{1}{\sin^2(x)}$;
- if $f(x) = \cosh(x)$, then $f'(x) = \sinh(x)$;
- if $f(x) = \sinh(x)$, then $f'(x) = \cosh(x)$;
- if $f(x) = \tanh(x)$, then $f'(x) = \dfrac{1}{\cosh^2(x)}$;
- if $f(x) = \coth(x)$, then $f'(x) = -\dfrac{1}{\sinh^2(x)}$.

Theorem. ***(necessary condition for the extremal values)***
Let $D \subset \mathbb{R}$ be nonempty open set, $f\colon D \to \mathbb{R}$ be function. If the function f has local extremum at point $x_0 \in D$ and f is differentiable at that point, then

$$f'(x_0) = 0.$$

Theorem. ***(Darboux)***
Let $f\colon (a,b) \to \mathbb{R}$ be differentiable function and $c, d \in (a,b)$, $c < d$. Then the derivative of function f takes all the value between $f'(c)$ and $f'(d)$ on the interval (c,d).

Theorem. ***(Cauchy)***
Let $f, g\colon [a,b] \to \mathbb{R}$ be continuous functions, suppose that they are differentiable on interval (a,b). Then there exists $\xi \in (a,b)$ such that

$$(f(b) - f(a))\, g'(\xi) = (g(b) - g(a))\, f'(\xi).$$

Theorem. ***(Lagrange)***
Let $f\colon [a,b] \to \mathbb{R}$ be continuous function, suppose that it is differentiable on interval (a,b). Then there exists $\xi \in (a,b)$ such that

$$f(b) - f(a) = (b-a) f'(\xi).$$

Theorem. ***(Rolle)***
Let $f\colon [a,b] \to \mathbb{R}$ be continuous function, suppose that it is differentiable on interval (a,b) and $f(a) = f(b)$. Then there exists $\xi \in (a,b)$ such that

$$f'(\xi) = 0.$$

Definition. *Let $D \subset \mathbb{R}$ be nonempty open set, $f\colon D \to \mathbb{R}$, $x_0 \in D$. If the function f is differentiable at a neighbourhood of x_0 and the derivative of function f is differentiable at point x_0, then we say that function f is twice differentiable at point $x_0 \in D$ and $(f')'(x_0)$ is called the second derivative of function f at point x_0 and we denote it with $f''(x_0)$.*

Let $n \in \mathbb{N}$. If the function f is differentiable n-times at a neighbourhood of x_0 and the nth derivative of function f is differentiable at point x_0, then we say that function f is $n+1$-times differentiable at point $x_0 \in D$ and $(f^{(n)})'(x_0)$ is called the $(n+1)$. derivative of function f at point x_0 and we denote it with $f^{(n+1)}(x_0)$.

Theorem.* *(Taylor)

Let $n \in \mathbb{N} \cup \{0\}$, $f\colon (a,b) \to \mathbb{R}$ and $x_0 \in (a,b)$. If the function f is differentiable $(n+1)$-times, then for all $x \in (a,b)$, $x \neq x_0$ there exists ξ between x and x_0 such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}.$$

The above polinomial

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

is called the nth Taylor polinomial of function f at x_0.

This formula

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$$

is called as Taylor-formula, and

$$\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$$

is called the remainder term.

Theorem.* *(L'Hospital rule)

Let $f, g\colon (a, a+r) \to \mathbb{R}$ be differentiable functions such that

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$$

and $g(x) \cdot g'(x) \neq 0$. If the limit

$$\lim_{x\to a} \frac{f'(x)}{g'(x)}$$

exists, then the limit

$$\lim_{x\to a} \frac{f(x)}{g(x)}$$

exists as well and they are equal.

Remark.

- *Similar definition is true for functions defined on $(a-r, a)$ or on $(a-r, a+r) \setminus \{a\}$.*
- *If the domains of f and g are not bounded above (or below), then the L'Hospital rule is valid with limits taken at the infinity.*
- *The L'Hospital rule is valid if*

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = \infty.$$

- *If*

$$\lim_{x\to a} f(x) = 0 \qquad \lim_{x\to a} g(x) = \infty$$

  then since

$$f(x) \cdot g(x) = \frac{f(x)}{\frac{1}{g(x)}}$$

  we can use the L'Hospital rule.

Definition. *The function* $f\colon (a,b) \to \mathbb{R}$ *is increasing on the interval* (a,b)*, if for all* $x, y \in (a,b)$*,* $x \le y$*:*

$$f(x) \le f(y).$$

The function $f\colon (a,b) \to \mathbb{R}$ *is decreasing on interval* (a,b)*, if for all* $x, y \in (a,b)$*,* $x \le y$*:*

$$f(x) \ge f(y).$$

If the above inequalities are strict for all $x \neq y$*, then we say that the function is strictly increasing/decreasing.*
It the function is either increasing or decreasing, then we say it is monotonic.

Theorem. (necessary condition for monotonicity)
Let $f\colon (a,b) \to \mathbb{R}$ *be differentiable function on* (a,b)*. Then* f *is increasing (decreasing) on* (a,b) *if and only if for all* $x \in (a,b)$ $f'(x) \ge 0$ *(*$f'(x) \le 0$*) holds.*

Theorem. (sufficient condition for extremal values)
Let $f\colon (x_0 - r, x_0 + r) \to \mathbb{R}$ *be differentiable function.*
If $f'(x) \ge 0$ *when* $x \in (x_0 - r, x_0)$ *and* $f'(x) \le 0$ *when* $x \in (x_0, x_0 + r)$*, then* f *has a local maximum at* x_0*.*
If $f'(x) \le 0$ *when* $x \in (x_0 - r, x_0)$ *and* $f'(x) \ge 0$ *when* $x \in (x_0, x_0 + r)$*, then* f *has a local minimum at* x_0*.*

Theorem. (general condition for extremal values)
Let $n \in \mathbb{N}$*,* $n \ge 2$*,* $f\colon (a,b) \to \mathbb{R}$ *be a function which is differentiable* n*-times at the point* $x_0 \in (a,b)$*. Suppose that*

$$f'(x_0) = f''(x_0) = \ldots = f^{(n-1)}(x_0) = 0, \quad \text{and} \quad f^{(n)}(x_0) \neq 0.$$

If

- n *is odd, then at the point* x_0 *there is no local extremum*
- n *is even and*
  - $f^{(n)}(x_0) > 0$*, then at point* x_0 *function* f *has local minimum;*
  - $f^{(n)}(x_0) < 0$*, then at point* x_0 *function* f *has local maximum.*

Definition. *Let* $I \subset \mathbb{R}$ *be nonempty interval. The function* $f\colon I \to \mathbb{R}$ *is convex, if for all* $x, y \in I$ *and for all* $\lambda \in [0,1]$*:*

$$f\left(\lambda x + (1 - \lambda y)\right) \le \lambda f(x) + (1 - \lambda) f(y).$$

Let $I \subset \mathbb{R}$ *be nonempty interval. The function* $f\colon I \to \mathbb{R}$ *is concave, if for all* $x, y \in I$ *and for all* $\lambda \in [0,1]$*:*

$$f\left(\lambda x + (1 - \lambda y)\right) \ge \lambda f(x) + (1 - \lambda) f(y).$$

Theorem. *The differentiable function* $f\colon I \to \mathbb{R}$ *is convex (concave) if and only if the function* $f'\colon I \to \mathbb{R}$ *is increasing (decreasing).*

Theorem. *Let* $f\colon I \to \mathbb{R}$ *be twice differentiable function.* f *is convex if and only if* $f''(x) \ge 0$ *holds for all* $x \in I$*. Moreover* f *is concave if and only if* $f''(x) \le 0$ *holds for all* $x \in I$*.*

Definition. *The value of the function at inner point x from I (where $f\colon I \to \mathbb{R}$) is a point of inflection if there exists $r > 0$, such that f is convex (or concave) on interval $(x-r, x)$ and f is concave (or convex) on interval $(x, x+r)$.*

Theorem. ***(condition for the point of inflection)***
If $f\colon I \to \mathbb{R}$ is differentiable function, then $(x, f(x))$ is a (inner) point of inflection if and only if it is an extremal value of f'.

Definition. *Let $f : (a,b) \to \mathbb{R}$ be a function. The differentiable function $F : (a,b) \to \mathbb{R}$ is called the indefinite integral (or antiderivative) of f, if $F' = f$. We denote it with $\int f$.*

Theorem. *If $f, F : (a,b) \to \mathbb{R},\ F' = f\ (F = \int f)$, then $G : (a,b) \to \mathbb{R}$ is an indefinite integral of f if and only if $\exists\, C \in \mathbb{R}$ such that $G(x) = F(x) + C$.*

Basic integrals:

$$\int \frac{1}{x}\, dx = \begin{cases} \ln(x) + C_1 & (x > 0) \\ \ln(-x) + C_2 & (x < 0) \end{cases}$$

$$\int x^\mu\, dx = \frac{x^{\mu+1}}{\mu+1} + C \qquad (x \in \mathbb{R}_+,\ \mu \neq -1)$$

$$\int a^x\, dx = \frac{a^x}{\ln a} + C \qquad (x \in \mathbb{R},\ a > 0,\ a \neq 1)$$

$$\int \sin(x)\, dx = -\cos(x) + C \qquad (x \in \mathbb{R})$$

$$\int \cos(x)\, dx = \sin(x) + C \qquad (x \in \mathbb{R})$$

$$\int \frac{1}{\sin^2(x)}\, dx = -\operatorname{ctg}(x) + C_k \qquad (x \in \,]\,k\pi, (k+1)\pi\,[\,,\ k \in \mathbb{Z})$$

$$\int \frac{1}{\cos^2(x)}\, dx = \operatorname{tg}(x) + C_k \qquad (x \in \,]\,k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}\,[\,,\ k \in \mathbb{Z})$$

$$\int \frac{1}{\sqrt{1-x^2}}\, dx = \arcsin(x) + C \qquad (x \in \,]-1, 1\,[\,)$$

$$\int \frac{1}{1+x^2}\, dx = \operatorname{arctg}(x) + C \qquad (x \in \mathbb{R})$$

$$\int \operatorname{sh}(x)\, dx = \operatorname{ch}(x) + C \qquad (x \in \mathbb{R})$$

$$\int \operatorname{ch}(x)\, dx = \operatorname{sh}(x) + C \qquad (x \in \mathbb{R})$$

$$\int \frac{1}{\sqrt{x^2+1}}\,dx = \operatorname{arsh}(x) + C = \ln(x+\sqrt{x^2+1}) + C \qquad (x \in \mathbb{R})$$

$$\int \frac{1}{\sqrt{x^2-1}}\,dx = \operatorname{arch}(x) + C = \ln(x+\sqrt{x^2-1}) + C \qquad (x \in]\,1, \infty\,[\,)$$

$$\int \left(\sum_{n=0}^{\infty} a_n x^n \right) dx = \sum_{n=0}^{\infty} a_n \frac{x^{n+1}}{n+1} + C \qquad (x \in]-\varrho, \varrho\,[\,)$$

Theorem. ***(linearity of the indefinite integral)***
Let $f, g : (a,b) \to \mathbb{R}$ be functions with existing $\int f$ and $\int g$, and $p, q \in \mathbb{R}$ are arbitrary, then there exists the integral $\int (pf+qg)$ and $C \in \mathbb{R}$ such that

$$\int [pf(x) + qg(x)]\,dx = p\int f(x)\,dx + q\int g(x)\,dx + C \qquad (x \in (a,b))\,.$$

Theorem. ***(integration by parts – indefinite integrals)***
It the functions $f, g : (a,b) \to \mathbb{R}$ are differentiable on (a,b) and there exists $\int f'g$, then $\int fg'$ exists as well and there exists $C \in \mathbb{R}$ such that

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx + C \qquad (x \in (a,b))$$

Theorem. ***(integration by substitution – indefinite integrals)***
Let $f : (a,b) \to \mathbb{R}$, $g : (c,d) \to (a,b)$ be functions with existing $g' : (c,d) \to \mathbb{R}$ and there exists $\int f$, then there exist $\int (f \circ g) \cdot g'$ and $C \in \mathbb{R}$ such that

$$\int f(g(x)) \cdot g'(x)\,dx = \left(\left(\int f \right) \circ g \right)(x) + C = \int f(t)\,dt\,\Big|_{t=g(x)} + C$$

$(x \in (c,d))$.

Remark. *If g^{-1} exists as well, then:*

$$\int f(x)\,dx = \left[\left(\int (f \circ g)g' \right) \circ g^{-1} \right](x) + C =$$
$$= \int f(g(t))g'(t)dt|_{t=g^{-1}(x)} + C$$

$(x \in (c,d))$.

Theorem. ***(integral and operations)*** *Let $f : (a,b) \to \mathbb{R}$ be differentiable on (a,b), $\alpha \in \mathbb{R} \setminus \{-1\}$, then the indefinite integral of the function $f^\alpha \cdot f'$ exists and there exists constant $C \in \mathbb{R}$ such that*

$$\int f^\alpha(x) \cdot f'(x)\,dx = \frac{f^{\alpha+1}(x)}{\alpha+1} + C.$$

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$, $f(x) \neq 0$ $(x \in [a,b])$, let f be differentiable on (a,b), then the indefinite integral of the function $\frac{f'}{f}$ exists and there exists constant $C \in \mathbb{R}$ such that

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C.$$

Let $f : \mathbb{R} \to \mathbb{R}$ be function, $\alpha, \beta \in \mathbb{R}$, $\alpha \neq 0$ be arbitrary. If $\int f(x)\,dx$ exists, then $\int f(\alpha x + \beta)\,dx$ exists as well and there exists $C \in \mathbb{R}$ such that

$$\int f(\alpha x + \beta)\,dx = \frac{F(\alpha x + \beta)}{\alpha} + C,$$

where F is the indefinite integral of f.

Let $[a,b] \subset \mathbb{R}$ be a closed interval. From now on we study bounded functions $f : [a,b] \to \mathbb{R}$.

Definition.

$$P = \{x_i \mid a = x_0 < x_1 < \cdots < x_i < \cdots < x_n = b\} \subset [a,b]$$

is called the partition of the interval $[a,b]$, the intervals $[x_{i-1}, x_i]$ $(i = 1, \ldots, n)$ are called subintervals of the partition. If $\Delta x_i = x_i - x_{i-1}$ then

$$\|P\| \doteq \sup\{\Delta x_i \mid i = 1, \ldots, n\}$$

is called the norm of the partition.

Definition. *Let P_1 and $P_2 \subset [a,b]$ be two partition. P_2 is a refinement of P_1, if $P_1 \subset P_2$.*

Definition. *Let (P_k) be a sequence of partitions of $[a,b]$ such that $\lim\limits_{k\to\infty} \|P_k\| = 0$.*

Definition. *Let $f : [a,b] \to \mathbb{R}$ be bounded function, P be a partition of $[a,b]$.*

$$M_i \doteq \sup_{x \in [x_{i-1}, x_i]} f(x), \qquad m_i \doteq \inf_{x \in [x_{i-1}, x_i]} f(x) \qquad (M_i, m_i \ \exists \text{ and } \in \mathbb{R})$$

Definition. *Let $f : [a,b] \to \mathbb{R}$ be bounded function, P be a partition of $[a,b]$. The numbers*

$$s(f,P) = \sum_{i=1}^{n} m_i \Delta x_i,$$

$$S(f,P) = \sum_{i=1}^{n} M_i \Delta x_i,$$

are called the lower and upper Darboux sum. If $t_i \in [x_{i-1}, x_i]$, then

$$\sigma(f,P) = \sum_{i=1}^{n} f(t_i) \Delta x_i$$

is called the Riemann sum.

Definition. *Let $f : [a, b] \to \mathbb{R}$ be bounded function. The numbers*

$$\underline{I} = \underline{\int_a^b} f \doteq \sup_P \{s(f, P)\}, \qquad \bar{I} = \overline{\int_a^b} f \doteq \inf_P \{S(f, P)\}$$

are called the lower and upper Darboux integral of f on $[a, b]$.

Theorem. *Let $f : [a, b] \to \mathbb{R}$ be bounded function, then $\underline{I}, \bar{I} \in \mathbb{R}$ and $\underline{I} \le \bar{I}$.*

Definition. *The bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$, if $\underline{I} = \bar{I}$. This value is called the Riemann (or definitive) integral of f on $[a, b]$ and it is denoted by $\int_a^b f$ or $\int_a^b f(x)\, dx$.*

Theorem. *If $f : [a, b] \to \mathbb{R}$ is continuous, then it is Riemann integrable.*

Theorem. *If $f : [a, b] \to \mathbb{R}$ is monotonic, then it is Riemann integrable.*

Theorem. ***(linearity of Riemann integral)***
If $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable, $p, q \in \mathbb{R}$, then $(p \cdot f + q \cdot g) : [a, b] \to \mathbb{R}$ is Riemann integrable and

$$\int_a^b (p \cdot f + q \cdot g) = p \cdot \int_a^b f + q \cdot \int_a^b g$$

Theorem. ***(Riemann integral and operations)*** *If $f : [a, b] \to \mathbb{R}$ is Riemann integrable, then f^2 is Riemann integrable,moreover if there exists $c > 0$ such that $|f(x)| \ge c$ for all $x \in [a, b]$, then $\dfrac{1}{f}$ is Riemann integrable.*
If $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable functions, then $f \cdot g$ is Riemann integrable, moreover if there exists $c > 0$ such that $|g(x)| > c$ for all $x \in [a, b]$, then $\frac{f}{g}$ is Riemann integrable.
If $f : [a, b] \to \mathbb{R}$ is Riemann integrable, then $|f|$ is Riemann integrable as well.

Theorem. ***(Newton-Leibniz formula)***
Let $f, F : [a, b] \to \mathbb{R}$ be functions, where f is Riemann integrable, F is continuous on $[a, b]$ and differentiable on (a, b), moreover $F'(x) = f(x)$ $(x \in (a, b))$, then

$$\int_a^b f = F(b) - F(a)$$

The number $F(b) - F(a)$ is often denoted with the symbol $[F(x)]_a^b$.

Theorem. ***(integration by parts - Riemann integral)***

If the functions $f, g : [a, b] \to \mathbb{R}$ are differentiable and their derivatives are continuous, then

$$\int\limits_a^b fg' = f(b)g(b) - f(a)g(a) - \int\limits_a^b f'g \,.$$

Theorem. ***(integration by substitution - Riemann integral)***

If the function $g : [a, b] \to [c, d]$ is differentiable and its derivative is continuous, function $f : [c, d] \to \mathbb{R}$ is continuous, then

$$\int\limits_a^b f(g(x))g'(x)\, dx = \int\limits_{g(a)}^{g(b)} f(x)\, dx \,.$$

Definition. *Let $a \in \mathbb{R},\ a < b \leq +\infty,\ f : [a, b) \to \mathbb{R}$ be bounded and Riemann-integrable function on all subinterval $[a, t] \subset [a, b)$. Suppose that either $b = +\infty$ or $\exists\, \varepsilon > 0$ such that f is not bounded on interval $[b - \varepsilon, b)$. If the (finite) limit*

$$\lim_{t \to b-0} \int\limits_a^t f \doteq \int\limits_a^b f$$

exists, then we call it improper Riemann integral of function f on $[a, b)$. In this case we say that $\int\limits_a^b f$ is convergent. (Otherwise divergent.)

Remark. *We can define improper integral with the setting: $a \in \mathbb{R},\ -\infty \leq c < a,\ f : (c, a] \to \mathbb{R}$ or $-\infty \leq a < b \leq +\infty$, $f : (a, b) \to \mathbb{R}$.*